



University Of Colombo, Sri Lanka

University of Colombo School of Computing

BACHELOR OF SCIENCE IN COMPUTER SCIENCE

First Year Examination – Semester I – UCSC AY21 [held in October 2024]

SCS1305 – Computer Systems

(Two (2) Hours)

Answer ALL questions

Number of Pages = 12

Number of Questions = 4

To be completed by the candidate

Index No:

Important Instructions to candidates:

1. The medium of instructions and questions is **English**.
2. Note that questions appear on both sides of the paper.
3. If a page or part of this paper is not printed, please inform the supervisor immediately.
4. Write your index number on every page of the Question paper.
5. Answer **ALL** questions.
6. There are **04** questions in **12** pages.
7. All **04** questions carry **equal marks**.
8. Write your answers and index number **clearly and readable** manner.
9. In questions where justifications, reasons, or steps of calculations are specifically asked, the student must provide the same in the answer. **Failure to provide** the above as required in the question will result in that question or part being awarded **zero marks**.
10. Any electronic device capable of storing and retrieving text including electronic dictionaries and mobile phones are **not allowed**.
11. Calculators are **NOT allowed**.

For Examiner's use only

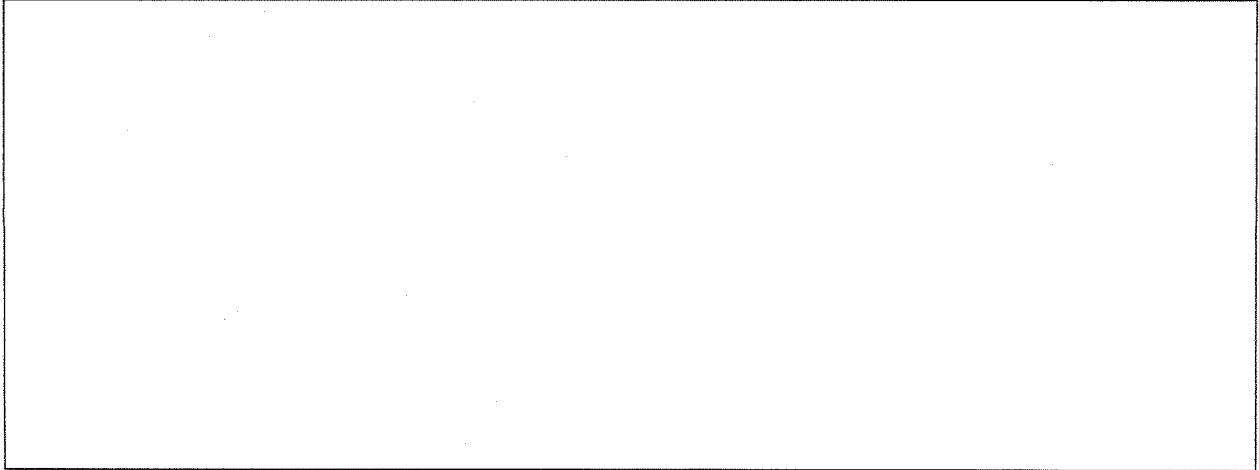
Question No	Marks
1	
2	
3	
4	
Total	

Index No:

Question 1

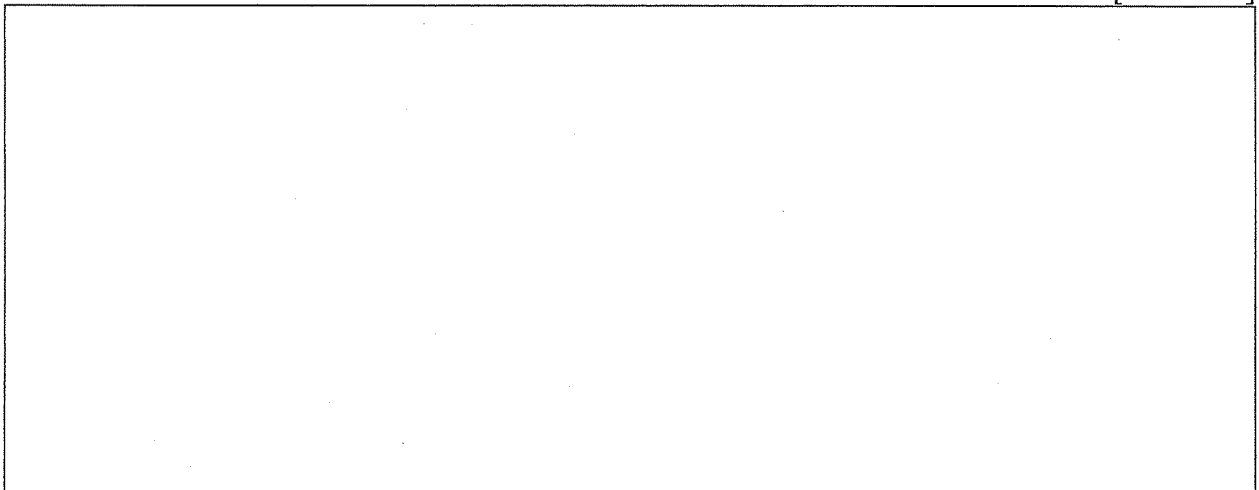
- A. What is the **8-bit signed magnitude** representation of **decimal number -126**. Show your calculations and steps.

[03 Marks]



- B. What is the **four digit hexadecimal** representation of the **decimal number 2540**? Show your intermediate steps clearly.

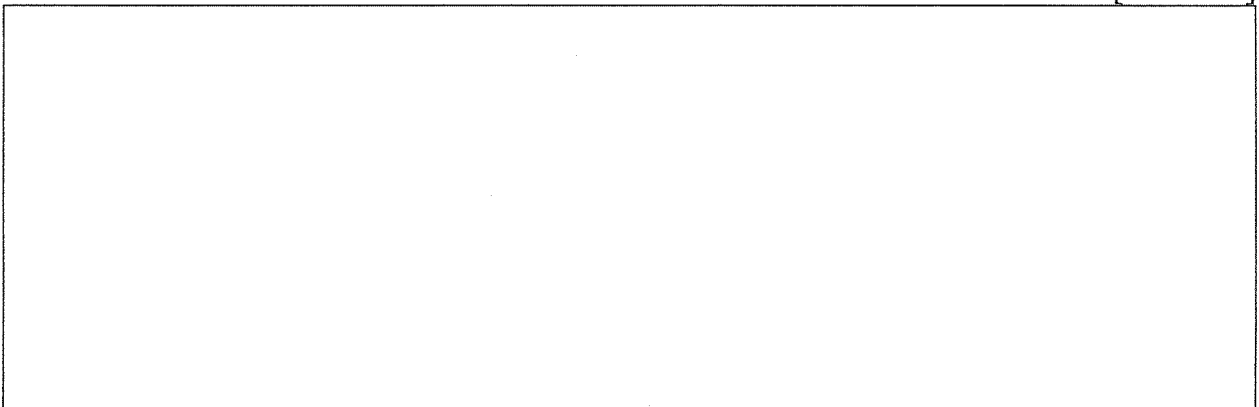
[04 Marks]



- C. What is the **decimal number** represented in the following **8-bit two's complement** number?

1011 1101

[04 Marks]



Index No:

- D. What are the lowest and highest values in **8-bit two's complement** representation? State both the binary representations and their corresponding decimal values.

[04 Marks]

- E. What is the **8-bit Excess-127** representation for **decimal** value **-126**? Show your steps in finalizing the answer.

[04 Marks]

- F. In IEEE 754 single-precision format 1 bit is for the sign, 8 bits are for the exponent, and 23 bits are for the significand (mantissa). What is the **decimal** number represented in following IEEE 754 representation? Show your calculations and steps clearly.

1 10000001 10110000000000000000000

[06 Marks]

Question 2

- A. Simplify the following Boolean function F and derive the simplest expression for F using Boolean algebraic laws, rules, and theorems. Show intermediate steps clearly.

$$F(A, B, C) = (A.B' + A'.B).(A' + B).(A + C)$$

[04 Marks]

- B. Derive the simplified Boolean algebraic expression in **Sum of Products** form for the following K-Map. Show your grouping clearly.

[03 Marks]

	$a.b$	$a.b'$	$a'.b'$	$a'.b$
$c.d$	0	X	0	1
$c.d'$	1	1	0	1
$c'.d'$	1	0	X	1
$c'.d$	1	0	0	0

- C. Derive the simplified Boolean algebraic expression in **Product of Sums** form for the following K-Map. Show your grouping clearly.

[03 Marks]

	$a.b$	$a.b'$	$a'.b'$	$a'.b$
$c.d$	0	X	0	1
$c.d'$	1	1	0	1
$c'.d'$	1	0	X	1
$c'.d$	1	0	0	0

- D. Assume you are tasked with designing a digital address decoding circuit for a memory system in a microcontroller-based embedded system. The memory system has eight (08) memory chips, and you need to select the appropriate memory chip based on the address provided by the microcontroller. Your goal is to design a circuit that enables the selection of one of memory chips given the right address.

What is the most appropriate common combinational logic circuit that can be used achieve this goal? State your reasoning for the answer.

[03 Marks]

- E. What is the minimum number of address lines required in the circuit above in the **Question 2.D**? Justify your answer.

[03 Marks]

- F. In the scenario given in **Question 2.D**, where you need to route data from the selected memory chip to the system's data bus, which common combinational logic circuit is most appropriate for this task? State your reasoning for the answer.

[03 Marks]

- G. Consider the sequential logic circuit given in Figure 2.1, which consist of a **negative edge triggered** D Flip Flop and a **positive edge triggered** J-K Flip Flop. The clock input is given as 'Clk' in the diagram.

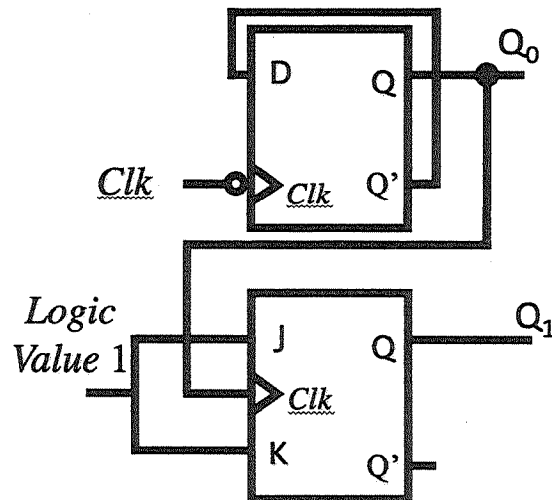
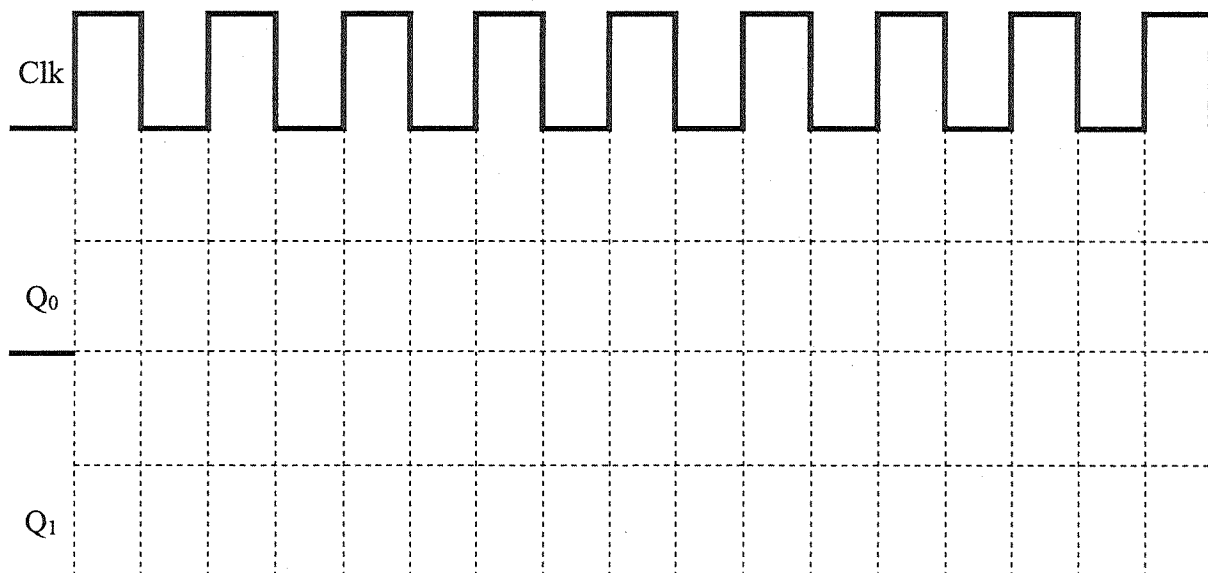


Figure 2.1 – The circuit diagram of the sequential logic circuit.

Complete the timing diagram given below for the above sequential logic circuit. Assume both Q_0 and Q_1 are in logic value zero (0) at the beginning as given in the timing diagram.

[06 Marks]



Question 3

A. Name which component of a CPU matches with each of the below given descriptions.

[08 Marks]

#	Description	CPU Component
i.	It decodes the instructions fetched from memory, translating them into signals that direct the operation of the CPU's various components.	
ii.	It provides extremely fast, temporary storage which is used to hold intermediate values and operands during computational operations, allowing the CPU to function in high speed.	
iii.	It holds the current instruction being executed, allowing the relevant components of the CPU to decode and execute it.	
iv.	It plays a critical role in managing the control flow of a program by directing the CPU to the correct instruction address to be executed next.	
v.	It supports a wide range of operations defined by the CPU's instruction set architecture (ISA), making it versatile and adaptable to different types of computational tasks.	
vi.	It leverages temporal locality and spatial locality to predict and prefetch data efficiently.	
vii.	It generates a regular, periodic signal that synchronizes all the operations within the CPU, ensuring that instructions and data transfers occur at precise intervals.	
viii.	It also handles bitwise operations, allowing for manipulation of individual bits within data words, which is critical for low-level programming and optimization.	

B. A compiler writer must choose between three code sequences *Seq.1*, *Seq.2*, and *Seq.3* for a certain high level language statement. Required instruction counts from three different instruction types A, B, and C for each of the three sequences are given in the Table 3.1 and their *Cycles Per Instruction* (CPI) are given in Table 3.2.

Table 3.1

	Instruction count per Type		
	A	B	C
<i>Seq.1</i>	2	1	2
<i>Seq.2</i>	3	2	1
<i>Seq.3</i>	2	2	1

Table 3.2

CPIs given in the Hardware Spec	
Instruction Type	CPI per instruction type
A	1
B	2
C	3

Index No:

Which sequence executes more instructions? Show your calculations and steps clearly.

[03 Marks]

C. Which sequence is faster in the scenario given in **Question 3.B**? Show your calculations and steps clearly.

[04 Marks]

D. Consider a machine architecture with an instruction set defined as follows.

LD R A	<i>Load the register R with the content of memory cell with the address A</i>
LI R I	<i>Load the register R with the value I in hexadecimal</i>
LR R1 R2	<i>Load the register R1 with the content of the register R2</i>
ST R A	<i>Store the content of register R to the memory cell whose address is A</i>
ADD R0 R1 R2	<i>Add the two integers in registers R1 and R2 and place the result in R0.</i>
SUB R0 R1 R2	<i>Subtract the integer in register R2 from the integer in register R1 and place the result in R0.</i>
MUL R0 R1 R2	<i>Multiply the integer in register R1 by the integer in register R2 and place the result in R0.</i>
JMP R A	<i>Jump to the instruction located in the memory cell with the address A if the bit pattern in R is equal to R0</i>
HLT	<i>Halt the execution</i>

An instruction layout in this machine architecture is **16 bits** long and **4 bits** of an instruction are allocated for **OPCODE**. Memory addresses are **8 bits** long and a register identifier is also **4 bits** long. All general-purpose registers are initialized to **0x00** at the time of a program starts its execution.

The following program is executed in this machine. Leftmost column in this set of instructions is the memory addresses that they are stored.

0x40.	LI	R1	0x05	
0x42.	LI	R2	0x01	
0x44.	LI	R3	0x01	
0x46.	SUB	R4	R1	R3
0x48.	JMP	R4	0x50	
0x4A.	MUL	R2	R2	R3
0x4C.	ADD	R3	R3	0x01
0x4E.	JMP	R0	0x46	
0x50.	ST	R2	0x80	
0x52.	HLT			

- i. State the values assigned to registers **R2**, **R3**, and **R4** in the order they are assigned during the execution of this program

[10 Marks]

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Question 4

A. In a hypothetical computer system with a *4-way set associative* cache and a *byte addressable* main memory, a cache line size is four (04) bytes. Three (03) bits of a memory address is reserved for the set index of the cache. Given this configuration, the CPU of this system referred the following list of addresses in a sequence and all of them resulted cache misses. The cache system follows, '*Least Recently Used*' cache replacement policy where it replaces the cache block that is least recently used whenever a replacement requirement arises.

- 1000 1001
- 0100 1010
- 0011 0110
- 0000 1011
- 0111 0001
- 0010 1000

i. What is the size of a single set in this cache system? Show your calculations and steps clearly. [02 Marks]

ii. What is the volume of this cache memory (size of the cache)? Show your calculations and steps clearly. [03 Marks]

- iii. Immediately after referring the above list of addresses, the CPU refers the following list of addresses in a sequence. State whether each of these address references would result a cache hit or cache miss with reasons.

- 0100 1011
- 0000 1010
- 0110 1001

[06 Marks]

- iv. Immediately after referring to the list of addresses given in Question 4.A.i, the CPU refers the following list of addresses in a sequence. State whether each of these address references would result a cache hit or cache miss with reasons.

- 1000 1001
- 0010 1000

[06 Marks]

Index No:

B. Consider a computer system with a *direct-mapped cache* that can hold a maximum of 64 cache lines (blocks), with each cache line capable of storing 16 bytes. The system has a main memory size of 4 GB and is *byte addressable*.

- i. Calculate the minimum address length required for this system. Show your calculations and steps clearly.

[02 Marks]



- ii. Determine the number of bits required for the offset, index, and tag fields in the cache addressing scheme. Show your calculations and steps clearly.

[06 Marks]

